

Zero-shot held-out (SSIM / power-weighted DoA azimuth error) — train 2 scenes, test on the unseen 3rd

Method	Conditioning	CBDcrossroad (in-dist)	parkinglot	ringroad (OOD)
Absolute (memorizes)	point xyz + tx_pos	0.14 / —	—	0.29 / —
Relative (ours)	single-bounce geom	0.86 / 10.6°	0.79 / 26.5°	0.70 / 83.0°
Mean-spectrum baseline	(no learning)	0.89 / 53.6°	0.95 / 20.9°	0.88 / 53.5°

Lower DoA error = predicted energy reaches the right Tx-dependent direction. SSIM saturates (baseline already 0.88–0.95); DoA discriminates.